Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions



Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data points per transaction

37

Missing Values

Review ratings imputed

Customer Data

- Demographics (Age, Gender, Location)
- Subscription Status
- Previous Purchases

Purchase Details

- Item, Category, Amount
- Season, Size, Color
- Discount & Promo Codes

Behavior Metrics

- Frequency of Purchases
- Review Ratings
- Shipping Type

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Data Preparation Process

01

Data Loading

Imported dataset using pandas

02

Exploration

Checked structure with info() and describe()

03

Missing Data

Imputed 37 missing review ratings by category

04

Standardization

Renamed columns to snake case format

05

Feature Engineering

Created age groups and purchase frequency days

06

Database Load

Connected to PostgreSQL for SQL analysis

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Revenue by Gender



Male customers generated **2.1x more revenue** than female customers

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Top Products by Rating

1

Gloves

3.86 average rating

2

Sandals

3.84 average rating

3

Boots

3.82 average rating

4

Hat

3.80 average rating

5

Skirt

3.78 average rating

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Shipping & Subscription Impact

Shipping Type

- **Standard:** $58.46 avg
- **Express:** $60.48 avg

Express shipping customers spend **$2 more** per purchase

Subscription Status

- **Subscribers:** 1,053 customers
- **Non-Subscribers:** 2,847 customers

Non-subscribers generate **2.7x more revenue**



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Discount-Dependent Products



50%

Hat

Half purchased with discount



49.66%

Sneakers

High discount dependency



49.07%

Coat

Seasonal discount pattern



48.17%

Sweater

Regular discount usage



47.37%

Pants

Discount-driven purchases

Top 5 products with highest percentage of discounted purchases

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Customer Segmentation

Loyal

3,116 customers

High repeat purchase rate

Returning

701 customers

Multiple purchases

New

83 customers

Single purchase only

98% of customers are either loyal or returning buyers

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Revenue by Age Group



Young adults drive the highest revenue, followed by middle-aged customers

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Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to increase subscriber base



Loyalty Programs

Reward repeat buyers to move them into loyal segment



Review Discount Policy

Balance sales boosts with margin control



Product Positioning

Highlight top-rated and best-selling products in campaigns



Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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